

# William Jackson

Senior Software Engineer – Ruby on Rails React AWS/Azure/GCP

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## PROFESSIONAL SUMMARY

Senior Software Engineer with 10+ years of experience delivering customer-facing SaaS platforms, internal business systems, and API-driven products across the **software**, **digital services**, and **platform engineering** industries, with deep specialization in **Ruby 2.3–3.2**, **Ruby on Rails 4.2–7.0**, **React 16–18**, scalable backend architecture, and production-grade frontend modernization. Strong record of building reliable monolith-to-service transition strategies, resolving high-impact performance defects, reducing page latency, improving release stability, and leading delivery of new product capabilities with secure REST APIs, background jobs, CI/CD automation, and cloud deployment practices that supported fast iteration without sacrificing maintainability.

## SKILLS

- **Frontend: React 16–18, JavaScript (ES6+), TypeScript**, Redux, Context API, Hooks, HTML5, CSS3, Sass, Responsive UI, Webpack, Vite, Jest, React Testing Library
- **Backend & APIs: Ruby 2.3–3.2, Ruby on Rails 4.2–7.0**, REST APIs, Sidekiq, Resque, Active Record, Service Objects, MVC Architecture, Background Jobs, Authentication, Authorization
- **Data & Messaging: PostgreSQL**, MySQL, Redis, Elasticsearch, SQL optimization, query tuning, caching strategies, asynchronous processing
- **Cloud & DevOps: AWS**, EC2, S3, RDS, CloudFront, Docker, Nginx, GitHub Actions, Jenkins, CI/CD pipelines, Linux, deployment automation
- **Observability, Quality & Security: RSpec**, Capybara, Jest, logging, performance profiling, code review, RBAC, audit trails, secure session handling, error monitoring, test automation

## WORK HISTORY

### Senior Software Engineer

February 2021 to December 2025

#### Viario Networks Inc. – (Delaware, United States)

- Architected and delivered a multi-tenant business platform using **Ruby on Rails 6.0–7.0** and **React 17–18**, separating domain logic into service layers that improved maintainability and reduced change failure rate by **38%** across major releases.
- Led the implementation of a new **React** administration experience for account provisioning, reporting, and workflow approvals, replacing server-rendered screens with reusable component patterns that improved page interaction speed by **35%**.
- Designed and exposed secure REST endpoints in **Rails** for billing, customer lifecycle events, and reporting workflows, enabling frontend and partner teams to consume a consistent contract while reducing integration defects by **30%**.
- Resolved a recurring production issue caused by N+1 queries and inefficient eager loading in **Active Record**, then introduced query profiling and indexing changes that cut heavy-report execution time from minutes to seconds.
- Migrated a legacy UI from jQuery-driven templates to **React 17–18** incrementally, using coexistence patterns and shared API contracts so feature delivery could continue without a risky big-bang rewrite.
- Implemented **Sidekiq**-based background processing for exports, notifications, and reconciliation tasks, which stabilized request latency under peak load and improved completion reliability for long-running jobs by **45%**.
- Introduced centralized authorization and audit logging in **Rails**, giving support and compliance teams clearer traceability across administrative actions while reducing security review findings in internal assessments.
- Worked closely with product managers and designers to deliver new subscription and self-service capabilities, translating ambiguous workflow requirements into shippable backend and frontend milestones with predictable delivery.
- Eliminated a class of intermittent session and token refresh issues affecting authenticated **React** flows by refactoring API error handling and retry logic, which reduced user-reported login failures by **50%**.
- Containerized core services with **Docker** and improved deployment automation through CI/CD workflows, shortening release preparation effort and making rollback procedures far more repeatable during production incidents.

- Improved test coverage across **RSpec**, request specs, and frontend unit tests, focusing on the most failure-prone modules first and helping the team raise release confidence for payment and workflow-heavy features.
- Mentored engineers on **Rails** service design, React component boundaries, and performance debugging, establishing practical review standards that improved code consistency across both new modules and migrated legacy areas.

**Software Engineer**  
**Avantia, Inc. – (Ohio, United States)**

October 2018 to January 2021

- Built core customer and operations features with **Ruby on Rails 5.2–6.0** and **React 16–17**, helping modernize a transactional web platform that required stable delivery across both legacy and newly introduced product areas.
- Implemented reusable **React** components for dashboards, form-heavy workflows, and status tracking experiences, reducing duplicated UI logic and improving frontend delivery speed for new features by **25%**.
- Created and maintained **Rails** APIs and server-side services for order management, notifications, and role-based workflows, with strong emphasis on predictable validations and clear separation of business rules.
- Solved recurring slowdowns caused by oversized serialized payloads and over-fetching on list views, then introduced pagination, selective serializers, and Redis-backed caching that materially improved response times.
- Upgraded portions of the application from older Rails conventions to cleaner service-oriented patterns, reducing controller complexity and making feature flags and phased rollout strategies easier to support in production.
- Integrated **Sidekiq** workers for async processing of emails, imports, and scheduled tasks, preventing request blocking during high-volume operations and improving system throughput during weekday peaks by **32%**.
- Partnered with QA and support teams to identify root causes of production defects, especially issues around stale cache data and race conditions, then shipped targeted fixes with regression tests to prevent recurrence.
- Refined database design and SQL queries in **PostgreSQL** for high-traffic screens, adding indexes and adjusting joins that reduced database load while keeping reporting accuracy intact for business users.
- Helped migrate frontend pages from mixed ERB and JavaScript behavior into clearer **React** modules, enabling more predictable state handling and reducing UI regressions in complex multi-step forms.
- Contributed to CI/CD improvements with automated test gates and deployment checks, which reduced manual verification work and improved release consistency for backend and frontend changes across environments.
- Collaborated in backlog planning and technical breakdown sessions, giving product stakeholders realistic implementation paths while balancing urgent bug resolution against longer-term modernization work.

**Software Developer**  
**ArnAmy, Inc. – (Florida, United States)**

March 2016 to September 2018

- Developed customer-facing and internal workflow features using **Ruby on Rails 4.2–5.2**, server-rendered views, and progressively introduced **React 16** where richer client-side interactivity created clear product value.
- Maintained and extended a monolithic **Rails** application with heavy business logic, gradually extracting reusable services and presenters so the codebase became easier to test, debug, and modify safely.
- Implemented account management, reporting, and notification functions while working closely with senior engineers to keep feature delivery aligned with operational needs and evolving customer expectations.
- Diagnosed and fixed several production issues tied to callback-heavy model behavior in **Rails**, replacing fragile logic with explicit service flows that reduced hidden side effects and support escalations.
- Introduced a lightweight **React** layer for selected dashboards and dynamic forms, improving usability without forcing a full frontend rewrite and giving the team a practical path toward modern UI development.
- Improved background processing reliability by restructuring delayed jobs, retry handling, and failure logging, which made support investigations faster and reduced silent job loss in business-critical workflows.
- Optimized SQL queries and cleaned up poorly indexed tables in **PostgreSQL** and MySQL-backed modules, improving report generation times and reducing lock contention during end-of-month processing.
- Added automated test coverage with **RSpec** and integration testing for the most defect-prone workflows, helping the team catch regressions earlier and reduce hotfix pressure after deployments.

- Supported application deployments, environment configuration, and production troubleshooting in Linux-based environments, building stronger operational awareness around logs, restarts, and release validation.
- Worked on incremental modernization efforts that prepared older modules for later upgrades to newer **Rails** and **React** patterns, keeping the application stable while still moving the platform forward.

**Junior Software Developer**  
**Centric Tech Inc. – (New Jersey, United States)**

October 2014 to February 2016

- Supported feature development and maintenance for internal business applications built with **Ruby on Rails 4.2–5.0**, working on CRUD workflows, validations, reporting screens, and small but important usability fixes.
- Built and maintained server-rendered views, form submissions, and backend actions in **Rails**, learning to handle production issues carefully while following established review and release practices.
- Investigated bugs related to inconsistent form state, invalid params handling, and fragile SQL queries, then delivered targeted fixes that improved data accuracy and reduced repeat tickets from internal users.
- Assisted with database schema changes, seed data updates, and basic query tuning in **PostgreSQL** and MySQL environments, gaining hands-on experience with how backend changes affected production stability.
- Contributed early frontend improvements through JavaScript and later small **React 16** components, especially in screens that needed better client-side validation or more responsive user feedback.
- Wrote unit and integration tests under guidance from senior engineers, helping strengthen release quality and building a solid foundation in test-driven maintenance of legacy-heavy applications.
- Participated in deployment support, log inspection, and issue triage for staging and production environments, which developed strong discipline around troubleshooting and safe rollout procedures.
- Learned to work effectively with Git-based collaboration, code reviews, and defect prioritization, steadily growing into larger ownership areas by pairing operational awareness with careful implementation.
- Improved older modules through refactoring of duplicated controller and helper logic, making future enhancements easier for the team and lowering the risk of regression in repeated business workflows.

## EDUCATION

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**Bachelor's degree in Computer Science**  
**Harvard University**

2010 to 2014

- Built a strong academic foundation in algorithms, databases, software engineering, and system design, then applied that knowledge through collaborative projects that mirrored real production development practices.

## CERTIFICATIONS

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- **AWS Certified Developer – Associate**  
Validates hands-on capability in building, deploying, and troubleshooting application workloads in AWS environments commonly used alongside modern **Rails** platforms.
- **Certified ScrumMaster (CSM)**  
Demonstrates practical understanding of agile delivery, iterative planning, and cross-functional collaboration that supports reliable execution in product engineering teams.
- **Redis for Developers Certification**  
Highlights applied knowledge of caching, session storage, and performance optimization patterns that are highly relevant to scalable **Ruby on Rails** applications.